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Transition-Metal-Free Cascade Enyne Rearrangement and Cyclopropanation of Allenylphosphine Oxides with *N*-Tosylhydrazones Accessing Alkynylcyclopropane Derivatives

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Abstract: We disclose an concise synthesis of alkynylcyclopropane derivatives containing adjacent quaternary carbon centers via a transition-metal free cascade enyne rearrangement and cyclopropanation process. This methodology, started from a variety of allenylphosphine oxides and N-tosylhydrazones, features the advantage of transition-metal-free, operational simplicity, wide substrate scope and good functional group tolerance.

Keywords: Cascade Enyne Rearrangement and Cyclopropanation; Alkynylcyclopropane; Transition-Metal-Free; Allenylphosphine Oxides; N-tosylhydrazones

Introduction

The efficient stereoselective synthesis of alkynylcyclopropane derivatives are of current interest since they play unique roles in biochemical processes, or display interesting pharmacological properties. Examples including **callipeltoside A** (cytotoxic activity against NSCLC–N6 human bronchopulmonary non-small-cell lung carcinoma and P388 cells),[1] **GT-2331** (protean agonist),[2] **efavirenz** (HIV-1 inhibitor),[3] and *ortho*-alkynylbenzoate (**ABz**, novel leaving group of glycosylation strategy)[4] are outlined in Figure 1.

Moreover, alkynyl cyclopropanes also constitute useful substrates for numerous chemical transformations[5] (including cycloaddition,[5a–d] ring expansion,[5e–i] rearrangement,[5j–k] C–H bond activation,[5l] and others[5m]). Thus, various approaches have been developed to prepare alkynylcyclopropanes, among which the functional conversion of alkynes is considered to be the most effective strategy (Scheme 1A).

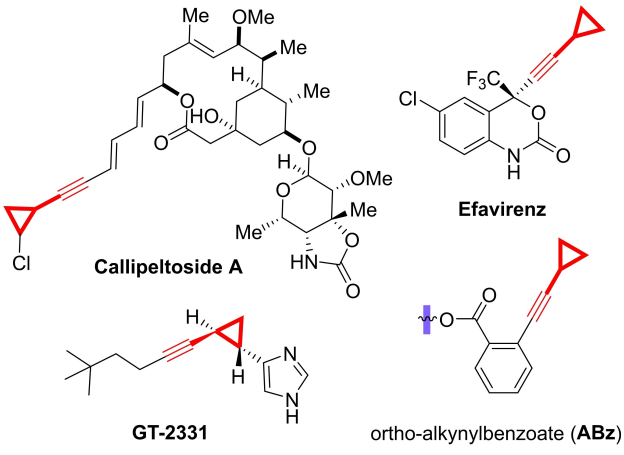
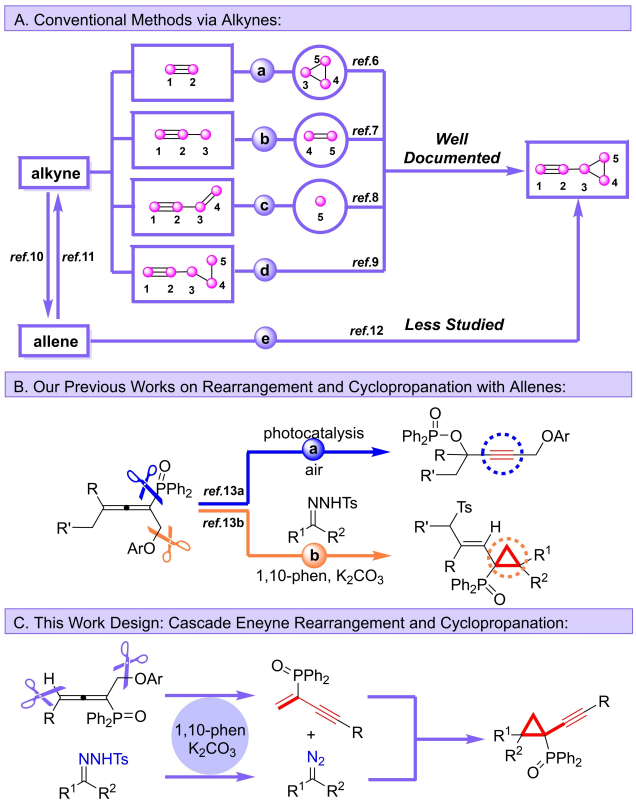


Figure 1. Compounds of medicinal interest containing alkynylcyclopropane scaffolds.

The strategies can be classified as follows: a) direct cyclopropanation of terminal alkynes with cyclopro-

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Scheme 1. (A) Current Synthetic Methods for Alkynylcyclopropanes; (B) Our Previous Works on Selective Rearrangement or Cyclopropanation of Allenylphosphine Oxides; (C) This Report.

pane sources (including methylenecyclo-propanes,[6a] cyclopropanol,[6b,c] cyclopropenes,[6d–g] and cyclopropyl iodides[6h–j]) in the presence of a transition metal; b) reactions between alkenes and alkynylcarbenes, which are generated from dihalogenated alkynes,[7a,b] alkynyldiazoacetates compounds,[7c–f] propargyl alcohols,[7g] and alkynyl Fischer-Carbene-Complexes (FCCs);[7h–l] c) [2 + 1] cycloaddition reactions of conjugated enynes with propargyl pivalate,[8a] *N*-tosylhydrazones,[8b–e] diazo compounds,[8f–l] (iodomethyl)-zinc species,[8m] or sulfur ylides;[8n] d) intramolecular cyclopropanation of functionalized alkynes.[9]

The conversion of alkynes to allenes[10]/allenes to alkynes[11] have been widely studied, however, most of the examples reported in the literature have been focused on the cyclopropanation of alkyne derivatives. To the best of our knowledge, allenes which act as alkyne surrogates have received considerably less attention on the synthesis of alkynylcyclopropanes. These limited reports are mainly focused on reactions *via* alkynyl carbene[12a,b] or intramolecular alkynyl cyclopropanation[12c–g] process. We envisioned that a reaction involving intermediate state of conjugated enyne may provide novel protocol to construct alkynyl cyclopropanes which has not been reported so far. This propose inspired us to design a new type of allenes to achieve this transformation. In this regime, recently, our group has advanced a series of studies on a kind of allenylphosphine oxides that promoted versatile synthetic transformations including intramolecular rearrangement and intermolecular cyclopropanation, respectively (Scheme 1B).[13a,b] As part of our continuing interest in allene and *N*-tosylhydrazone chemistry,[13] herein, we report for the first time, 1,10-Phen/K_2CO_3 promoted synthesis of phosphinyl alkynylcyclopropanes containing adjacent quaternary carbon centers based on the couplings of 1,1,3-trisubstituted allenylphosphine oxides with *N*-tosylhydrazones (Scheme 1C).

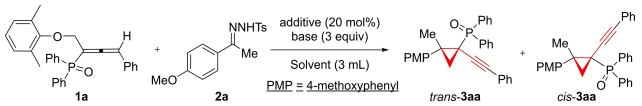
Results and Discussion

To rationalize our hypothesis, our discovery and initial development of this reaction is summarized in Table 1. The substrates of allenylphosphine oxide (**1 a**) and **2 a** were submitted to various conditions. The initial attempt utilizing a mixture of 3 equiv. of K_2CO_3 in 1,4-dioxane at 110 °C provided 1,1,2,2-tetrasubstituted alkynylcyclopropane **3 aa** in 65% isolated yield and 4.9:1 dr (*trans:cis*) (entry 1). Subsequent screening of bases and solvents revealed that K_2CO_3 was the best in terms of dr value (entries 1–6), and 1.4-dioxane was the optimal solvent in terms of both reaction efficiency and dr value (entries 7–14). In view of our previous work,[13b] we found that the addition of 20 mol% 1,10-phen resulted in a great increase in product yield (91% yield), albeit with a slightly diminished dr value after some trials with various additives (entries 15–20). Further evaluation of its loading (entries 21–23) led to lower yield. Moreover, control experiment revealed that the base was indispensable to obtain the product (entry 24). Therefore we choose the condition in entry 15 with the highest yield as the standard condition, although the *dr* value is not the highest.

With the optimal conditions in hand, the scope of the cascade enyne rearrangement and cyclo-propanation reaction was investigated. Structural variations on the alkynyl unit of the allenylphosphine oxides were examined (Scheme 2 **A**). All reactions smoothly took place under standard conditions to furnish alkynylcyclopropane **3 aa**–**3 ka** in good yields (57–91%). The relative stereochemistry of *trans*-isomer was assigned based on the X-ray crystallographic analysis of **3 ba** (CCDC: 2034008).[14] Furthermore, the position of the substituents (*para*, *meta*, or *ortho*) was found to have little impact on the efficiency of this transformation (**3 ca**, **3 ga**, and **3 ha**). As for the stereoselectivity, all of the substrates tested, with either electron-donating (4-MeO–Ph, 4-Me–Ph, 2-Naph) groups, alkyl groups (Et, Cy) or electron-withdrawing (4-CF_3–Ph, 4-F–Ph, 4-Cl–Ph) showed a slight trans-preference. Among which, 4-MeO–Ph on allenylphosphine oxides ren-

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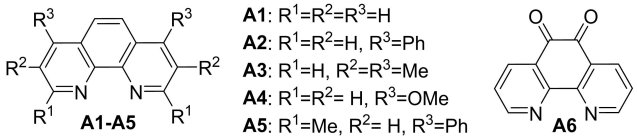
Table 1. Optimization of the Reaction Conditions.[a]



entry	additive	base	solvent	dr (*trans:cis*)[b]	yield **3aa** (%)[c]
1	–	K_2CO_3	1,4-dioxane	4.9:1	65
2	–	^tBuOK	1,4-dioxane	3.7:1	33
3	–	$KHCO_3$	1,4-dioxane	2.0:1	61
4	–	Na_2CO_3	1,4-dioxane	3.0:1	60
5	–	DBU	1,4-dioxane	2.9:1	40
6	–	DMAP	1,4-dioxane	3.3:1	33
7	–	K_2CO_3	toluene	3.4:1	42
8	–	K_2CO_3	THF	2.0:1	38
9	–	K_2CO_3	CH_3NO_2	–	0
10	–	K_2CO_3	H_2O	–	<5
11	–	K_2CO_3	DMF	3.3:1	42
12	–	K_2CO_3	DMSO	3.0:1	20
13	–	K_2CO_3	EtOH	3.6:1	30
14	–	K_2CO_3	DCE	3.0:1	13
15[d]	**A1**	**K_2CO_3**	**1,4-dioxane**	**3.0:1**	**91**
16[d]	**A2**	K_2CO_3	1,4-dioxane	3.2:1	70
17[d]	**A3**	K_2CO_3	1,4-dioxane	2.7:1	64
18[d]	**A4**	K_2CO_3	1,4-dioxane	2.7:1	67
19[d]	**A5**	K_2CO_3	1,4-dioxane	2.7:1	70
20[d]	**A6**	K_2CO_3	1,4-dioxane	2.9:1	80
21[e]	**A1**	K_2CO_3	1,4-dioxane	2.9:1	80
22[f]	**A1**	K_2CO_3	1,4-dioxane	2.8:1	72
23[g]	**A1**	K_2CO_3	1,4-dioxane	2.9:1	70
24[d]	**A1**	–	1,4-dioxane	–	0

[a] Reaction conditions: allenylphosphine oxide (**1a**, 0.3 mmol), *N*-tosylhydrazones (**2a**, 0.6 mmol), additive (20 mol%), base (3 equiv.), solvent (3 mL), N_2, 110 °C, 18 h;
[b] Diastereomeric ratios (dr = *trans*/*cis*) were determined by ^{31}P NMR spectroscopy of the products with TEPA (triethylphosphonoacetate) as an internal standard;
[c] Isolated yields;
[d] Additive (20 mol%) was used;
[e] Additive (10 mol%) was used;
[f] Additive (50 mol%) was used
[g] Additive (100 mol%) was used.

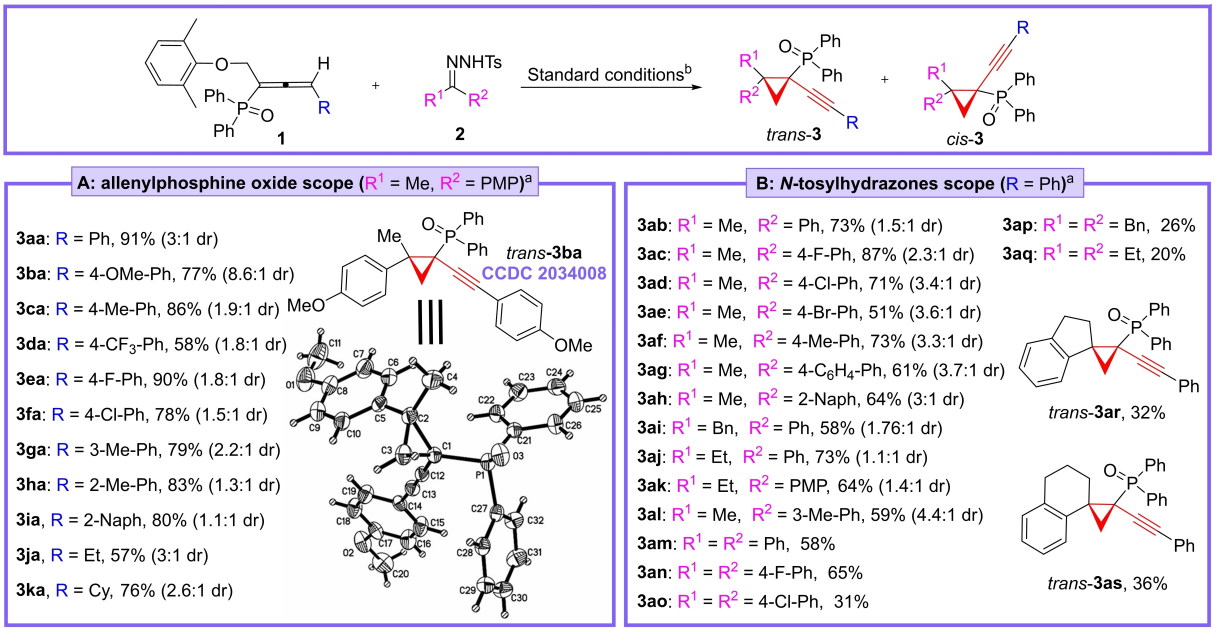
A1 = 1,10-phenanthroline; **A2** = Bathophenanthroline; **A3** = 3,4,7,8-Tetramethyl-1,10-phenanthroline; **A4** = 4,7-Dimethoxy-1,10-phenanthroline; **A5** = Bathocuproine; **A6** = 1,10-Phenanthroline-5,6-dione.



dered the corresponding product **3ba** with the highest *dr* value of 8.6:1.

The reactivity of a range of *N*-tosylhydrazones (**2**) was examined next (Scheme 2 **B**). The reaction proceeded smoothly with *N*-tosylhydrazones **2b**–**2k**, all of which delivered the desired tetra-substituted alkynylcyclopropane **3ab**–**3ak** in good yields (51–87%), although with slightly decreased *dr*. Electron-donating substituents (3-Me–Ph) at the *meta*-positions rendered the cascade enyne rearrangement and cyclopropanation reaction with a relative higher trans-selectivity (**3al**). For symmetric *N*-tosylhydrazones, we were pleased to find that diphenyl (**2m**), di-*p*-fluoroaryl (**2n**), di-*p*-chloroaryldiphenyl (**2o**), dibenzyl (**2p**), diethyl (**2q**) substituents provide the corresponding tetra-substituted alkynylcyclopropane **3am**, **3an**, **3ao**, **3ap**, and **3aq** with only one stereogenic center in 58%, 65%, 31%, 26% and 20% yields respectively. In addition, 1-indan (**2r**), 1-tetrahydronaphthalene (**2s**) substituents provided the potential chiral center spiro-

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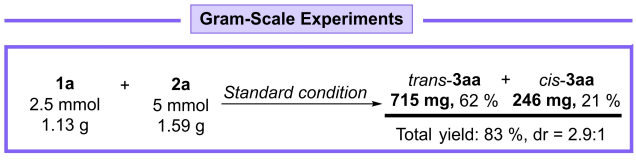


Scheme 2. Generality of the cascade enyne rearrangement and cyclopropanation reaction: (A) Allenylphosphine oxide Scope; (B) *N*-tosylhydrazones Scope. [a]Isolated yield by column chromatography; [b]Standard conditions: allenylphosphine oxide (**1a**, 0.2 mmol), *N*-tosylhydrazones (**2a**, 0.4 mmol), 1,10-Phen (20 mol%), K_2CO_3 (3 equiv.), N_2, 110 °C, 18 h. PMP = 4-methoxyphenyl.

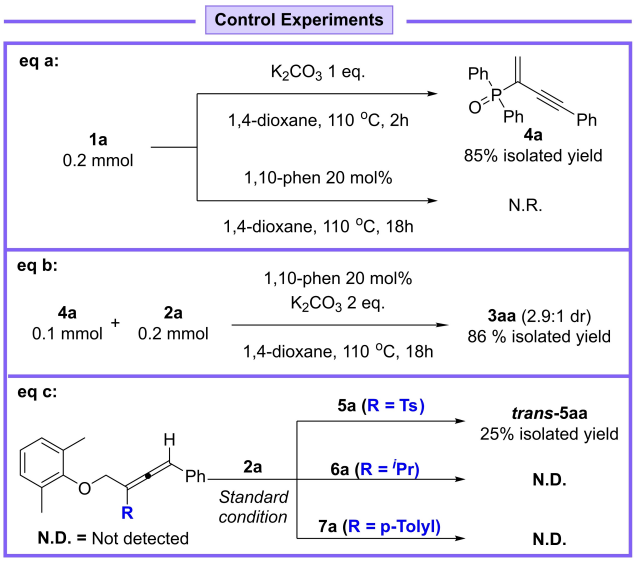
alkynylcyclopropane **3ar** and **3as** as a single diastereoisomer in 32 and 36% yields, respectively.

To verify the practical application of this enyne rearrangement and cyclopropanation reaction, the cascade reaction of **1a** and **2a** was conducted on the 2.5 mmol scale, and as expected, the corresponding products *trans*-**3aa** and *cis*-**3aa** were obtained in a yield of 62% and 21%, respectively (Scheme 3).

After exploring the scope of the reaction, we then performed some controlling experiments to study the mechanism of this transformation (Scheme 4). First, **4a** could be isolated in 85% isolated yield when K_2CO_3 (1 equiv.) were added to **1a** in 1,4-dioxane at 110 °C for 2 h, but no product could be obtained with 1,10-Phen (20 mol%) only (Scheme 4, **eq a**). It implies that base might play a crucial role in the formation of conjugated enyne. Then the reaction of **4a** and **2a** in the presence of 1,10-Phen (20 mol%) and K_2CO_3 (2 equiv.) at 110 °C for 18 h efficiently delivered the corresponding product in 86% yield with more or less the same *dr* value (Scheme 4, **eq b**), which confirmed that the reaction was achieved via intermediate **4a**.



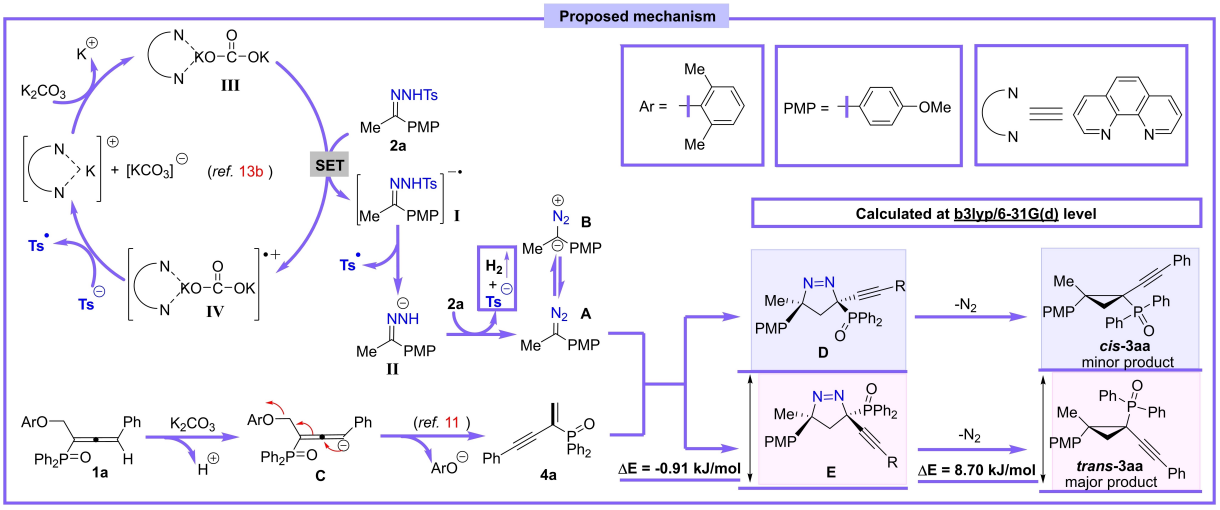
Scheme 3. Gram-Scale Experiments.



Scheme 4. Control Experiments.

With the consideration of clarifying the "Ph_2PO" species directed cyclization, analogues including tosyl-substituted allene (**5a**), isopropyl-substituted allene (**6a**) and *p*-tolyl-substituted allene (**7a**) were synthesized and applied to the cyclization conditions (Scheme 4, **eq c**). The allene with electron-withdrawing (Ts) group (**5a**) only delivered the corresponding *trans*-product in 25% yield. However, these allenes with electron-donating (^{i}Pr, *p*-tolyl) groups (**6a**, **7a**)

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Scheme 5. Proposed mechanism.

decomposed rapidly, without detection of any proposed corresponding products.

On the basis of these experimental results and previous reports,[11,13b,15] we tentatively proposed the reaction mechanisms outlined in Scheme 5. Initially, single electron transfer (SET) from the 1,10-phen/K_2CO_3 complex to *N*-tosylhydrazone **2 a** gives a radical anion **I**, which undergoes N–S bond cleavage to result in a tosyl radical and an *N*-anion intermediate **II**. The *N*-anion intermediate **II** will leave the hydrogen anion rapidly to give a diazo compound **A**.[13b] Tosyl anion and molecular hydrogen evolution comes from the deprotonation of **2 a** by the in situ generated hydrogen anion. And intermediate **4 a**[11] were gradually and sustainably formed under the treatment of K_2CO_3 from allenylphosphine oxide **1 a**. Isomerization of **A** afforded the zwitterionic intermediate **B**. Subsequent [3 + 2] cycloaddition of intermediate **A** or **B** with the intermediate **4 a** gives the pyrazoline **D** or **E**. However, theoretical calculations indicate that the energy difference between the intermediate **D** and the intermediate **E** ($\Delta E = E(\mathbf{D}) - E(\mathbf{E}) = -0.9$ kJ/mol) is almost negligible. After nitrogen is released, the energy of the product *cis*-**3 aa** is only 8.7 kJ/mol higher than that of the product *trans*-**3 aa**, which rationalized the *trans*-**3 aa** as the major product and the *cis*-**3 aa** as the minor product.[15]

Conclusion

In summary, we have reported a transition-metal-free cascade enyne rearrangement and cyclopropanation reaction between *N*-tosylhydrazones and allenylphosphine oxides. This protocol provides a straight way to afford the various adjacent quaternary carbon centers containing alkynylcyclopropane derivatives in high yields via a [3 + 2] cycloaddition process. The key step of the reaction depends on the formation of the intermediate conjugated enyne. Further investigations towards the synthetic applications of the alkynyl cyclopropanes are ongoing in our laboratory.

Experimental Section

General Procedure for the Transition-Metal-Free Cascade Enyne Rearrangement and Cyclopropanation of allenylphosphine oxide with N-Tosylhydrazones to Access compounds (3)

A 10 mL two-necked flask equipped with condenser and a stirring bar under nitrogen was charged with allenyl-phosphine oxide (**1**, 0.3 mmol, 1.0 equiv.), *N*-tosylhydrazones (**2**, 0.6 mmol, 2.0 equiv.), 1,10-Phen (11.9 mg, 20 mol%), K_2CO_3 (124 mg, 0.9 mmol, 3.0 equiv.) and 3 mL degassed 1,4-dioxane. The reaction mixture was then heated to 110 °C for 18 hrs in an oil bath until the complete consuming of **1** monitored by TLC. The reaction mixture was cooled to room temperature, after all of the volatiles were removed under vacuum, extracted with DCM and H_2O. The combined organic layers were dried over anhydrous Na_2SO_4 and evaporated in vacuo, the crude product was purified on flash chromatography on silica gel using petroleum ether/ethyl acetate/dichloromethane to afford product **3**.

Acknowledgements

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